

Article

Coordinated Optimization of Wind Farm Control Parameters for Primary Frequency Regulation Based on Fatigue Load Prediction

Maxin Sun, Yuqing Jin *  and Xiaohua Shi

School of Electrical and Power Engineering, Hohai University, Nanjing 211100, China;
231306080037@hhu.edu.cn (M.S.); 18115326615@163.com (X.S.)

* Correspondence: jyq16@hhu.edu.cn

Abstract

With the increasing penetration of wind power, the participation of wind farms in primary frequency regulation has become important for maintaining power system frequency stability. However, virtual inertia and droop control, while providing frequency support, can increase structural fatigue loads in wind turbines and shorten their service life. To address this issue, this study proposes a coordinated optimization method for wind farm primary frequency control parameters based on fatigue load prediction. First, damage equivalent load (DEL) data under different power disturbances, wind speeds, and control parameter settings are generated through OpenFAST–Simulink co-simulation. Then, a multilayer perceptron (MLP) neural network is developed to establish the mapping from power disturbance, wind speed, and control parameters to turbine DEL. Based on the trained model, an optimization framework is constructed to minimize the total DEL of the wind farm, improve the uniformity of DEL distribution among turbines, and satisfy grid frequency support constraints. Simulation results show that the proposed method effectively reduces the overall fatigue load of the wind farm while ensuring system frequency security and improving load distribution uniformity among turbines.

Keywords: wind turbines; primary frequency regulation; damage equivalent load (DEL); MLP neural network; control parameter optimization

1. Introduction

With the increasing penetration of renewable energy such as wind power in power systems, the proportion of traditional synchronous generators continues to decline, leading to a significant reduction in system equivalent inertia and increasingly prominent frequency stability issues [1,2]. Existing studies have shown that high penetration of renewable energy weakens system inertia support, making the system more prone to higher rates of change of frequency, lower frequency nadirs, and reduced frequency recovery capability under active power disturbances, thereby posing serious challenges to secure and stable power system operation [3–5]. To adapt to the operational requirements of low-inertia power systems, increasing research attention has been devoted to control methods enabling renewable energy units, especially wind turbines, to participate in system frequency support [6,7].

Currently, wind turbines mainly participate in grid frequency regulation through supplementary active-power control strategies. Existing methods can generally be classified into two categories according to the source of frequency-support energy and the associated control mechanism: one is frequency-support control based on the release of rotor kinetic energy, and the other is power-reserve control based on de-loaded operation to maintain



Academic Editor: Roberto Zivieri

Received: 28 March 2026

Revised: 23 April 2026

Accepted: 30 April 2026

Published: 2 May 2026

Copyright: © 2026 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article distributed under the terms and

conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

a power margin [8]. The former typically releases rotor kinetic energy in the initial stage of a sudden system frequency change through supplementary inertia response, virtual inertia control, or fast power regulation based on the rate of change of frequency (ROCOF), thereby improving the ROCOF and elevating the frequency nadir. Its advantages include a fast response speed, making it suitable for the rapid support needs of low-inertia systems, but it may also induce secondary power fluctuations during the rotor speed recovery phase. The latter continuously provides active power support after a frequency deviation occurs, typically through deloaded operation, reserving power margin, or coordination with energy storage. Its advantages are a longer support duration and relatively stable control, but it often requires sacrificing partial wind energy utilization efficiency. Extensive research has been conducted on both types of methods. Ref. [9] advanced from fixed-parameter control to multi-objective adaptive combined inertia-droop control, to enhance frequency response performance and improve the recovery characteristics of the unit. Ref. [10] proposes a closed-loop synthetic inertia control strategy for wind turbines combined with over-speeding de-loaded operation, which uses detected frequency deviation as a feedback signal, to reduce power imbalance during frequency regulation, and effectively decreases the maximum frequency deviation of the system. Overall, existing studies primarily evaluate the effectiveness of wind turbine frequency regulation control from the perspective of enhancing frequency response performance.

However, the participation of wind turbines in frequency regulation is not without associated costs. During frequency support, the unit must rapidly adjust its active power output in response to the system frequency deviation and the rate of change of frequency (ROCOF). These second-level power adjustments substantially exacerbate dynamic oscillations in key load paths, including drivetrain torque and tower fore–aft bending moments, thereby accelerating the fatigue damage accumulation of key components [11–14]. To address the structural fatigue issues arising from wind turbine participation in frequency regulation, research has been initiated to explore strategies such as auxiliary control optimization, mechanistic analysis of inertia control, and coordinated frequency support with energy storage systems [15–18]. Relevant research indicates that frequency support provided by wind turbines is often accompanied by structural fatigue costs. Suppressing fatigue damage while ensuring frequency regulation performance has become a critical issue in research on wind power frequency support.

At present, investigations into the coupled problem of frequency response and fatigue loads are predominantly conducted using high-fidelity aeroelastic simulation tools such as OpenFAST (v3.5.0) and Bladed (v4.8). In contrast to analytical approaches that focus exclusively on electrical power responses or utilize low-order simplified models, high-fidelity co-simulation platforms are capable of simultaneously capturing the multiphysics coupling characteristics across the aerodynamic, structural, drivetrain, and control subsystems. Consequently, they are better suited for the high-fidelity evaluation of critical fatigue load channels under diverse operating scenarios. Within this framework, OpenFAST—an open-source, high-fidelity aeroelastic simulation suite—boasts distinct advantages, including comprehensive modeling capabilities, robust modular coupling, and the precise characterization of dynamic responses for blades, towers, and drivetrain components. Accordingly, it has been extensively utilized in the domains of wind turbine load analysis and control strategy research. Nevertheless, they suffer from high modeling complexity, long simulation times, and substantial computational cost [19]. When frequency regulation control parameters require repeated evaluation across various disturbance levels, wind speeds, and parameter combinations, relying solely on high-fidelity co-simulation makes it difficult to meet the demands of rapid analysis and optimization.

By comparison, machine learning approaches offer novel insights into the rapid assessment of structural fatigue loads in wind turbines. In recent years, relevant studies have made substantial progress in the rapid modeling of wind turbine fatigue response. Ref. [20] established a data-driven surrogate model that maps wind input time series to DEL, which can replace some time-consuming aeroelastic simulations with high accuracy. Subsequent studies further combined SCADA measurements with open-source model data, utilizing Gaussian Process Regression (GPR) to achieve rapid DEL assessment for operating wind turbines [21]. Ref. [22] used neural networks to surrogate the rainflow-counting process, achieving near-real-time acceleration of fatigue life estimation. Ref. [23] applied intelligent learning methods to the active power regulation of wind turbines, considering both power command tracking and fatigue mitigation of the main shaft and tower. The aforementioned research demonstrates that machine learning techniques are capable of accurately capturing the nonlinear mapping relationships between variables such as wind speed, operating status, and fatigue loads. By serving as approximate surrogates for high-fidelity simulations, these approaches provide a robust and efficient tool for the rapid prediction of fatigue loads under complex operating scenarios.

However, most existing machine learning load surrogate models focus on power optimization and fatigue load assessment [24,25], whereas few studies have considered the direct influence of frequency-regulation control parameters on fatigue loads. In primary frequency response scenarios, these control parameters directly determine the intensity of frequency support, the power regulation process, and the resulting additional load levels. Therefore, surrogate models established under normal operating conditions are difficult to apply directly to rapid fatigue assessment under frequency-regulation conditions. Moreover, when the scope of analysis expands from individual turbines to the wind-farm level, the challenge is no longer limited to the rapid estimation of fatigue loads for a single turbine, but also involves integrating single-turbine fatigue-response assessment into the optimization of multi-turbine control parameters so as to support coordinated decision-making at the wind-farm level.

To address these issues, this paper proposes an optimization method for wind farm primary frequency control parameters based on fatigue load prediction. First, a co-simulation platform based on OpenFAST and MATLAB/Simulink is employed to obtain the structural load time series of wind turbines under different frequency-regulation control parameters and to calculate the corresponding damage equivalent loads (DELs). Then, a multilayer perceptron (MLP) neural network is developed to predict DEL values from power disturbance, wind speed, and control parameters, thereby enabling rapid fatigue-load assessment under frequency-regulation conditions. Finally, for multi-unit coordinated frequency-regulation scenarios in wind farms, a control-parameter optimization model is formulated by jointly considering total wind-farm fatigue, inter-unit load balancing, and system frequency security constraints. On this basis, a rolling optimization strategy is adopted to achieve differentiated coordination of virtual inertia and droop coefficients, thereby realizing coordinated optimization between system frequency-support performance and wind turbine fatigue loads. Furthermore, considering that the complete rolling optimization process is time-consuming and cannot directly satisfy real-time online solution requirements, a control strategy table for medium disturbance levels is constructed based on offline optimization results to enable rapid deployment of control parameters after disturbances, thereby balancing optimization performance and application efficiency. Compared with existing research, the main characteristics of this study are: first, performing rapid fatigue load assessment specifically for primary frequency control conditions; second, combining high-fidelity co-simulation with machine learning-based data-driven prediction to balance load analysis accuracy and parameter optimization efficiency; and third, further extending

DEL prediction to the multi-turbine collaborative optimization problem of wind farms to achieve a coordinated trade-off between frequency support performance and fatigue load levels.

The remainder of this paper is organized as follows: Section 2 describes the wind turbine model and the principles of primary frequency control. Section 3 establishes the co-simulation platform based on OpenFAST and MATLAB/Simulink to generate DEL sample datasets across various operating scenarios. Subsequently, an MLP neural network-based DEL prediction model is developed and validated using these datasets. Section 4 proposes a rolling optimization framework for the primary frequency control parameters of wind farms. Section 5 presents the wind farm simulation setup and discusses the experimental results. Section 6 provides a comprehensive discussion of the methodology, and Section 7 concludes the study with suggestions for future research directions.

2. Wind Turbine Model and Frequency Regulation Method

2.1. Wind Turbine Model

A wind turbine is a system that converts wind energy into mechanical energy through its mechanical structure and subsequently into electrical energy via a generator. It typically consists of an aerodynamic subsystem, drivetrain, pitch control system, electrical system, control subsystem, and tower.

The aerodynamic subsystem captures wind energy and mainly comprises the blades and hub, collectively referred to as the rotor. As wind passes through the rotor, it is decelerated, and part of its kinetic energy is converted into mechanical energy. In theoretical analysis, the key physical quantities used to describe the aerodynamic characteristics of a wind turbine are the mechanical power P_m and the aerodynamic thrust F_t which are expressed in Equations (1) and (2), respectively [26].

$$P_m = \frac{1}{2} \rho \pi R^2 C_p(\lambda, \beta) v^3 \quad (1)$$

$$F_t = \frac{1}{2} \rho \pi R^2 C_t(\lambda, \beta) v^2 \quad (2)$$

where ρ is the air density; R is the rotor radius, $C_p(\lambda, \beta)$ is the power coefficient, β is the pitch angle, v is the incoming wind speed, λ is the tip speed ratio defined as $\lambda = \omega_r R / v$, ω_r is the rotor speed and C_t is the thrust coefficient of the wind turbine.

In fatigue analysis of the tower structure, the fatigue load at the tower base is typically used as a representative indicator of the overall structural fatigue load.

Therefore, the bending moment at the tower base is adopted as a key indicator to evaluate tower fatigue loads [27], as expressed in Equation (3).

$$M_t = H_t F_t \quad (3)$$

where H_t is the height of the wind turbine tower.

The drivetrain is a critical mechanical structure that connects the aerodynamic system and the electrical system of a wind turbine, and its primary function is to transmit the mechanical energy captured by the rotor from the airflow to the generator via the main shaft and gearbox, enabling the conversion of mechanical energy into electrical energy.

In the fatigue load analysis of wind turbines, in order to more accurately describe the dynamic characteristics of the drivetrain, a two-mass model is adopted to represent the

drivetrain, which can capture the torsional dynamic behavior between the rotor side and the generator side, as expressed in Equation (4).

$$\begin{cases} J_t \frac{d\omega_r}{dt} = T_m - T_{sh} \\ J_g \frac{d\omega_g}{dt} = T_{sh} - T_g \\ T_{sh} = K_{sh}\theta_{tw} + D_{sh} \frac{d\theta_{tw}}{dt} \\ \frac{d\theta_{tw}}{dt} = \omega_r - \frac{\omega_g}{N} \end{cases} \quad (4)$$

where T_m is the input mechanical torque of the wind turbine, T_{sh} is the shaft torque, T_g is the electromagnetic torque of the generator, J_t is the inertia time constant of the wind turbine, J_g is the inertia time constant of the generator, ω_g is the synchronous speed of the generator, N is the gearbox ratio, K_{sh} is the stiffness coefficient of the drivetrain shaft, θ_{tw} is the torsional angle of the shaft and D_{sh} is the damping coefficient.

Considering that this study focuses on the frequency response process of wind turbines participating in primary frequency regulation, the time scale of interest is on the order of seconds; whereas the electromagnetic dynamics of the converter respond much faster, typically on the order of milliseconds, and can be regarded as fully settled relative to primary frequency regulation dynamics. Therefore, without compromising the accuracy of system frequency response analysis, the electromagnetic transient process of the converter is neglected, and a simplified generator model is adopted:

$$P_e = T_g \omega_g \quad (5)$$

where P_e is the output power of the wind turbine.

2.2. Frequency Regulation Method

During system power disturbances, wind turbines contribute to frequency regulation via virtual inertia and droop control strategies [8]. Virtual inertia control enables rapid release of turbine kinetic energy in the early stage of frequency variation, improving the system's inertial response capability. Droop control provides stable power support during the sustained frequency deviation stage. The corresponding power regulation amount ΔP can be expressed as shown in Equation (6).

$$\Delta P = -K_{df} \frac{df}{dt} - K_{pf} \Delta f \quad (6)$$

where K_{df} is the virtual inertia coefficient and K_{pf} is the droop coefficient.

3. Fatigue Load Data Acquisition and MLP Prediction Model

3.1. Fatigue Load Evaluation

In the below-rated wind speed region, wind turbines can participate in grid frequency regulation via coordinated virtual inertia control. Virtual inertia control primarily responds to the rate of change of frequency, whereas droop control responds to frequency deviation; their combined action enables rapid active power adjustment over short time scales. However, such power regulation induces dynamic variations in generator electromagnetic torque, drivetrain torque, and tower aerodynamic thrust, thereby modifying the load-cycle characteristics of critical structural components, including the main shaft and tower. Since fatigue damage is primarily determined by the amplitude and frequency distribution of load cycles, the participation of wind turbines in frequency regulation can significantly influence the process of structural fatigue accumulation.

To quantify this effect, the rainflow counting method is first employed to decompose the load time series into individual load cycles. This method identifies equivalent cycles

from irregular variable-amplitude load histories and determines the amplitude and number of occurrences of each cycle, thus providing the basis for subsequent fatigue damage assessment. The material S–N curve is then used to establish the relationship between load amplitude and fatigue life, which describes the number of cycles that the material can withstand under a given stress or load level [28]. On this basis, Miner’s linear damage accumulation rule is adopted to calculate the cumulative fatigue damage caused by load cycles with different amplitudes, thereby obtaining the total fatigue damage over the reference period. Finally, the accumulated damage is converted into the damage equivalent load (DEL), which is used to characterize the fatigue effect of loads on critical structural components of wind turbines. The 1 Hz DEL corresponding to the reference period T_{ref} is calculated as follows:

$$DEL = \left(\sum \frac{N_i S_i^m}{N_{ref}} \right)^{\frac{1}{m}} \quad (7)$$

where N_i is the number of cycles in the i -th load interval (obtained by rainflow counting), S_i is the load amplitude of the i -th cycle obtained from rainflow counting, m is the slope exponent of the material S–N curve in logarithmic coordinates which depends on the material properties of the structural component and N_{ref} is the reference number of cycles, typically taken as the equivalent number of cycles over the design lifetime of the wind turbine.

In this study, the tower base fore–aft bending moment, tower base side-to-side bending moment, and rotor torque are selected as the primary fatigue load channels. The selection is motivated by the fact that the tower base fore–aft and side-to-side bending moments represent the loading characteristics of the tower in different directions, whereas the rotor torque captures load variations in the drivetrain; these variables correspond to the tower and drivetrain, which are the critical structural components most significantly influenced by frequency regulation control [11,29]. In contrast, blade loads are more sensitive to stochastic wind speed fluctuations and exhibit relatively weak responses to changes in control parameters under the primary frequency regulation scenarios considered in this study [30]. Therefore, the above three load channels are selected for fatigue analysis, and their detailed information is listed in Table 1.

Table 1. Information for each load channel.

Load Channels	Description	Unit
LSShftMxa	Rotor Torque	[kN·m]
TwrBsMx	Tower Base Lateral Bending Moment (also referred to as side-to-side bending moment)	[kN·m]
TwrBsMy	Tower Base Fore-Aft Bending Moment (also referred to as fore-aft bending moment)	[kN·m]

3.2. Impact of Frequency Regulation Actions on Wind Turbine

OpenFAST is a modular, open-source wind turbine dynamics simulation tool developed by the National Renewable Energy Laboratory (NREL), which has been certified by Germanischer Lloyd (GL) and can accurately simulate the dynamic response of both rigid and flexible mechanical structures of wind turbines [31].

To comprehensively simulate the frequency response of wind turbines in power system frequency regulation, while accounting for mechanical structural dynamics and enabling

fatigue load analysis, a co-simulation model of the NREL 5 MW turbine is established using OpenFAST and Matlab/Simulink (v2025a). The turbine mechanical dynamics are modeled in OpenFAST, whereas the wind turbine control system [32] and a fast grid simulation model (GSFR) based on the IEEE 9-bus system are developed in Matlab/Simulink, and deep integration between OpenFAST and Simulink is realized through an S-Function interface. The IEEE 9-bus system considered has a total load of 692 MW. The detailed co-simulation model is shown in Figure 1. Table 2 lists the basic parameters of the NREL 5 MW wind turbine [33].

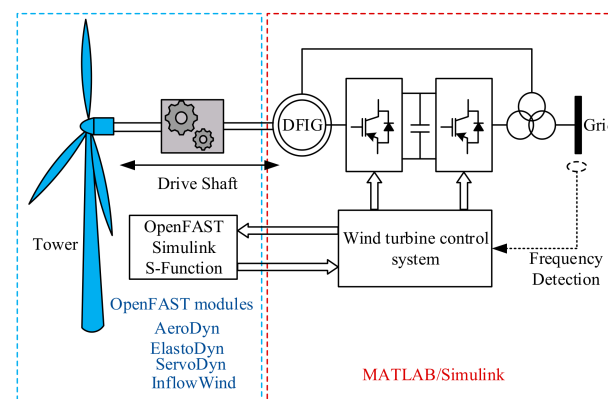


Figure 1. The detailed co-simulation model: the blue part represents the mechanical structural dynamic model of the wind turbine constructed in OpenFAST, and the red part represents the wind turbine control system and the IEEE 9-bus grid-connected model established in Matlab/Simulink.

Table 2. The basic parameters of the NREL 5 MW wind turbine.

Parameter	Value	Parameter	Value
Rated power	5 MW	Number of blades	3
Rotor diameter	126 m	Cut-in wind speed	3 m/s
Hub height	3 m	Cut-out wind speed	25 m/s
Hub diameter	90 m	Rated wind speed	11.4 m/s

To improve simulation efficiency, the IEEE 9-bus system is simplified into a system frequency response (SFR) model, and the transfer function $G_{SFR}(s)$ is expressed as shown in Equation (8):

$$G_{SFR}(s) = \frac{-0.0384s^3 - 0.1057s^2 - 0.0169s - 3.77 \times 10^{-4}}{s^4 + 1.7054s^3 + 0.7290s^2 + 0.2276s + 0.0095} \quad (8)$$

The input is the system power imbalance:

$$P_{input} = \Delta P_L - P_{wf} \quad (9)$$

where ΔP_{sys} is the system power disturbance and P_{wf} is the wind farm frequency regulation power.

The variation in system frequency can be expressed as:

$$\Delta f(s) = G_{SFR}(s)P_{input}(s) \quad (10)$$

The frequency is given as:

$$f(s) = 50(1 + \Delta f(s)) \quad (11)$$

To investigate the influence of coordinated virtual inertia control on the mechanical loads of wind turbines, simulation analysis is conducted based on the previously established electromechanical coupled wind turbine model. Figure 2 shows the distribution of damage equivalent loads under different virtual inertia coefficients.

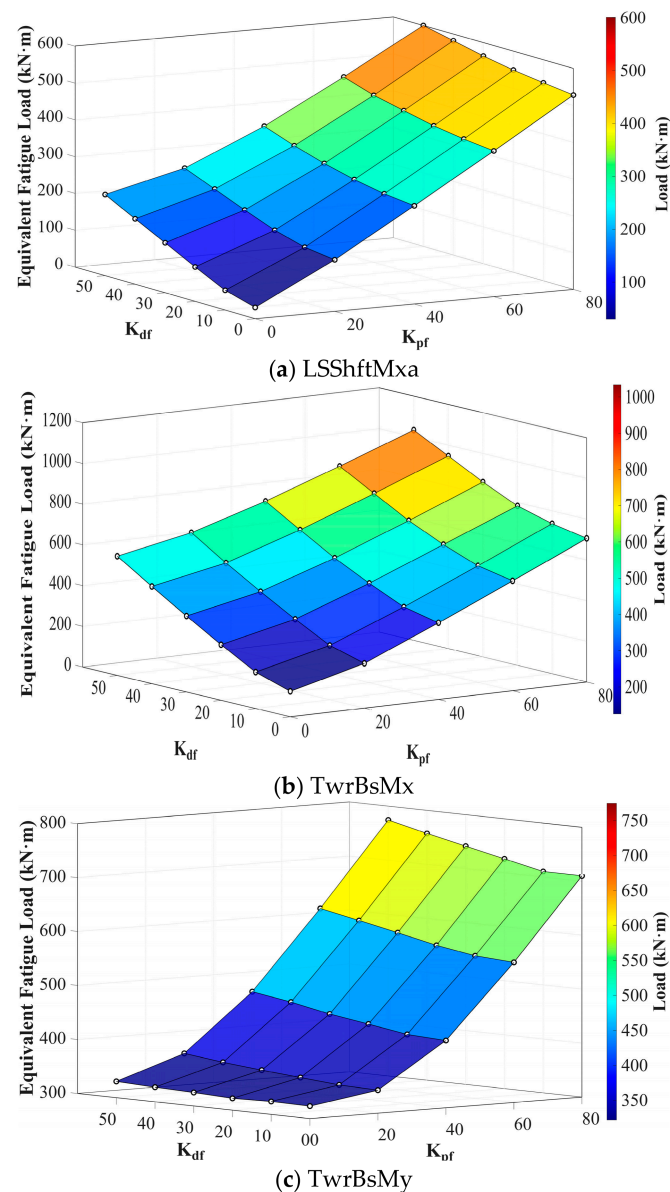


Figure 2. Distribution of damage equivalent loads under different virtual inertia coefficients.

From a structural response perspective, the additional power introduced by frequency regulation leads to an imbalance between mechanical torque and electromagnetic torque, thereby exacerbating fluctuations in drivetrain torque and tower bending moments. In other words, higher values of K_{df} and K_{pf} are associated with an increase in the overall DEL of the wind turbine, indicating a positive correlation between the control gains and structural fatigue load. Therefore, there exists an inherent trade-off between improved frequency regulation capability and increased structural fatigue. The key objective of this study is to, while satisfying frequency regulation requirements, effectively reduce the overall fatigue load of the wind farm and significantly improve the balance of load distribution among turbines, thereby avoiding the concentration of frequency regulation

burden on a small number of units, which could compromise the overall lifespan of the wind farm.

3.3. Wind Turbine Damage Equivalent Load Characteristic Database

Based on the theoretical calculation of DEL, the fluctuation of the main shaft torque originates from variations in wind resources and active power output. The fluctuation of the tower base bending moment is mainly related to the aerodynamic thrust acting on the turbine. Based on the above analysis, this study selects power disturbance magnitude, wind speed, virtual inertia coefficient, and droop coefficient as the fundamental operating parameters to characterize the DEL of the drivetrain and tower.

To ensure that the constructed sample scenarios have a clear system background and engineering representativeness, the selection of parameter ranges for the frequency regulation conditions in this paper is not arbitrarily set based on experience. Instead, it is determined based on the established IEEE 9-bus system model [34]. This system is composed of thermal power units, hydropower units, and a wind farm, which can reflect the typical frequency dynamic characteristics under the coexistence of traditional synchronous units and wind farms. Therefore, the representativeness of the operating scenarios discussed in this paper primarily targets the typical operation of wind farms participating in primary frequency regulation within such hybrid thermal–hydro–wind power systems.

The wind speed data samples used in this study were obtained from a real wind farm (hereinafter referred to as the LY wind farm), which consists of 48 DFIG units. The power output and wind speed data of all turbines in the LY wind farm for the entire year of 2015 were collected; however, because the cut-in wind speed of the DFIG model was 5 m/s, only data corresponding to wind speeds above 5 m/s were retained, with a sampling interval of 10 min. Since the original wind speed data were sampled at a minute-level time scale, whereas primary frequency regulation occurs on a second-level time scale, there exists a significant temporal mismatch between the two. On the second-level time scale, wind speed varies relatively slowly and can therefore be assumed constant during a single frequency regulation event. Based on this, a constant wind speed assumption was adopted when constructing individual frequency regulation samples, whereby the wind speed of each turbine within the corresponding time period was treated as a constant input parameter.

During the sample construction process, the range of power disturbance magnitudes is determined based on the frequency response results under the IEEE 9-bus system SFR model in this paper. Specifically, by analyzing the response law of the frequency nadir varying with disturbances, this paper selects a power disturbance interval that covers different levels of frequency support demand. This represents typical scenarios of wind farm participation in primary frequency regulation under varying disturbance intensities. On this basis, the power disturbance ranges from 0.0283 to 0.1133 with a step size of 0.0094. The wind speed ranges from 7 to 11 m/s with a step size of 0.5 m/s. Both the virtual inertia coefficient and droop coefficient range from 0 to 100 with a step size of 10. A total of 9801 frequency regulation scenarios are generated based on these parameter combinations.

In the co-simulation model, operating conditions with a simulation duration of 200 s were generated, and the load time series of two key components—the tower and drivetrain—under frequency support were obtained using OpenFAST. The DEL over this period was then calculated via rainflow counting, thereby establishing a DEL database for wind turbines during frequency regulation.

3.4. MLP Neural Network Prediction Model

Due to the complex nonlinear mapping among frequency regulation variables, operating conditions, and fatigue loads, as well as the high computational cost associated with directly using high-fidelity co-simulation for DEL evaluation during optimization, this study employed a multilayer perceptron (MLP) neural network to construct a fast DEL prediction model. The MLP is a typical feedforward neural network with strong nonlinear function approximation capability and can learn complex mappings between input variables and output responses through multiple hidden layers, making it suitable for fast regression modeling of the relationship between frequency regulation variables and fatigue load indicators of wind turbines [35]. In this study, the power disturbance magnitude, wind speed, virtual inertia coefficient, and droop coefficient in the sample database were used as inputs to the neural network, while the DEL values corresponding to the tower fore–aft bending moment, tower side-to-side bending moment, and rotor torque were used as outputs, enabling rapid prediction of key fatigue loads under frequency regulation conditions.

The prediction performance of the MLP neural network is relatively sensitive to hyperparameters such as the hidden layer structure, activation function, learning rate, regularization coefficient, and batch size. To reduce the empiricism in the hyperparameter selection process, this paper uses the grid search combined with 5-fold cross-validation method to optimize the key hyperparameters of the model. Specifically, different parameter combinations are trained and validated individually within the predefined search space, and the coefficient of determination (R^2) under 5-fold cross-validation is used as the primary evaluation metric to select the parameter configuration with the best comprehensive performance. Table 3 MLP model hyperparameter search space and optimal results. shows the hyperparameter search space and optimal results of the MLP model. The hyperparameter search results show that when the hidden layer structure is (64, 32), the activation function is ReLU, the learning rate is 0.001, the regularization coefficient is 0.001, and the batch size is 64, the model achieves a good balance among prediction accuracy, training stability, and complexity. Therefore, the subsequent model training and testing in this paper are all based on this optimal parameter configuration.

Table 3. MLP model hyperparameter search space and optimal results.

Hyperparameter	Search Space	Optimal Value
Hidden layer architecture	[(64,32), (128,64), (128,64,32), (256,128,64)]	(64,32)
Activation function	[ReLU, Tanh, Leaky ReLU]	relu
Learning rate	[0.001, 0.01, 0.1]	0.001
Regularization coefficient	[0.0001, 0.001, 0.01]	0.001
Batch size	[32, 64, 128]	64

The training and inference of the MLP model are implemented based on the PyTorch platform, and the samples are divided into training and testing sets at a ratio of 80% to 20%. Meanwhile, the input and output data are standardized, and an early stopping mechanism is introduced to improve the stability of model training. To reduce the risk of model overfitting, a Dropout mechanism is introduced in the hidden layers. The ReLU activation function is employed to bolster the network's nonlinear approximation capabilities. Finally, the Mean Squared Error (MSE) is adopted as the loss function, and the model is optimized utilizing the Adam optimizer.

To enhance the comprehensiveness of the model description, the mathematical formulation of the MLP network training process is further detailed. Let the input and output vectors be defined as follows:

$$\begin{aligned} x &= [\Delta P, v, K_{df}, K_{pf}]^T \\ y &= [DEL_{LSShftMx}, DEL_{TwrBsMx}, DEL_{TwrBsMy}]^T \end{aligned} \quad (12)$$

For the l -th layer, the forward propagation process is mathematically formulated as follows:

$$\begin{aligned} a^{(l)} &= f(z^{(l)}) \\ z^{(l)} &= W^{(l)}a^{(l-1)} + b^{(l)} \end{aligned} \quad (13)$$

where $W^{(l)}$ and $b^{(l)}$ are the weight matrix and bias vector of the l -th layer, respectively, $a^{(l-1)}$ is the output of the previous layer, and $f(\cdot)$ is the activation function.

In this study, the ReLU activation function is employed in the hidden layers, formulated as follows:

$$f(x) = \max(0, x) \quad (14)$$

To quantify the deviation between the predicted values and the ground-truth values, the Mean Squared Error (MSE) is adopted as the loss function:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_{i,pre} - y_{i,real}| \quad (15)$$

where $y_{i,pre}$ is the predicted value of the i -th data point, $y_{i,real}$ is the actual value of the i -th data point; and n is the total number of samples.

During the training process, the Adam optimization algorithm is utilized to update the network parameters. The iterative update process is mathematically expressed as:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \\ \hat{m}_t &= \frac{m_t}{1 - \beta_1^t}, \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \\ \theta_t &= \theta_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon} \end{aligned} \quad (16)$$

where $g_t = \partial L / \partial \theta_t$ is the gradient of the loss function with respect to the parameters, θ_t represents the parameters to be optimized, and η is the learning rate.

To evaluate the predictive performance of the constructed model, the coefficient of determination R^2 and the mean relative absolute error δ_m are adopted as evaluation metrics, and the computational efficiency of high-fidelity co-simulation and MLP-based fast prediction is further compared.

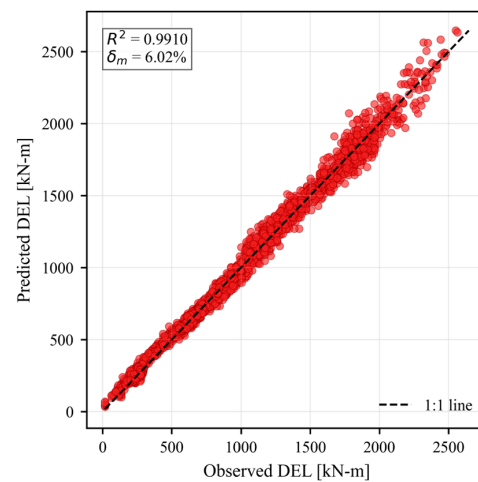
$$R^2 = 1 - \frac{\sum_{i=1}^n (y_{i,pre} - y_{i,real})^2}{\sum_{i=1}^n (y_{mean} - y_{i,real})^2} \quad (17)$$

$$\delta_m = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_{i,pre} - y_{i,real}}{y_{i,real}} \right| \times 100\% \quad (18)$$

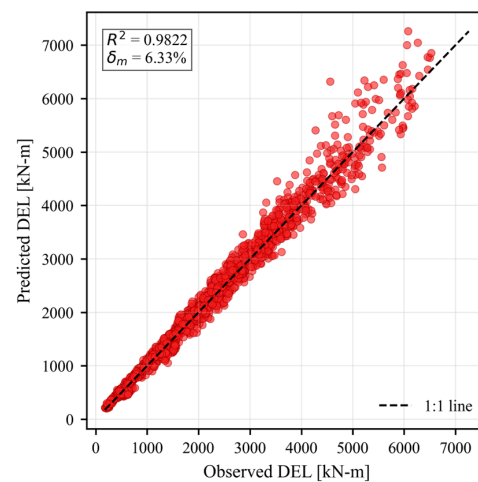
where $y_{i,pre}$ is the predicted value of the i -th data point, $y_{i,real}$ is the actual value of the i -th data point and y_{mean} is the mean of the actual data.

The prediction accuracy of the models for rotor torque and tower-induced bending moment load channels is shown in Figure 3 and Table 4. As shown in Table 4 and Figure 3, the R^2 values of the three load channels are all greater than 0.98, indicating a high level

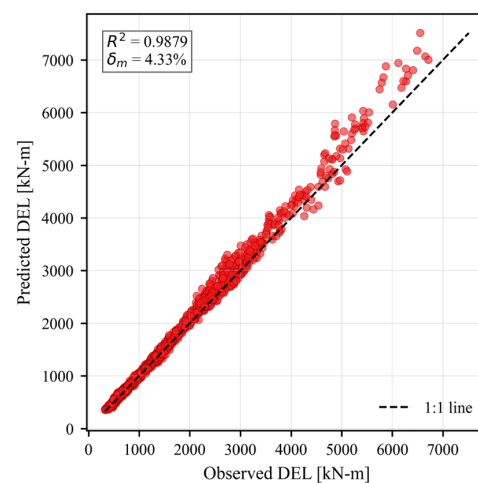
of fitting accuracy. The δ_m values are all below 7%, further validating the accuracy and stability of the model predictions. Based on the above evaluation metrics, the proposed MLP model exhibits strong predictive performance for fatigue load channels and can achieve high-accuracy DEL prediction.



(a) LSShftMxa



(b) TwrBsMx



(c) TwrBsMy

Figure 3. Scatter plots of predicted versus ground-truth values for individual load channels in the testing set.

Table 4. Performance metrics of the model in the test dataset.

Load Channels	$\delta_m/\%$	R^2
LSShftMxa	6.02	0.9910
TwrBsMx	6.33	0.9822
TwrBsMy	4.33	0.9879

To further verify the rationality of using the MLP model for DEL prediction, this paper compares it with several common machine learning methods, including Random Forest (RF), Decision Tree (DT), Gradient Boosting Decision Tree (GBDT), and Linear Regression (LR). The aforementioned models are trained and tested based on the same dataset split and input variables, and the coefficient of determination (R^2) and Mean Absolute Percentage Error (δ_m) are adopted as evaluation metrics. The comparison results are shown in Table 5.

Table 5. Comparison of prediction performance among different models.

Load Channel	Methods	R^2	$\delta_m/\%$
LSShftMxa	MLP	0.9910	6.02
	RF	0.9723	9.71
	DT	0.9716	9.81
	GBDT	0.9606	12.38
	LR	0.7814	31.01
TwrBsMy	MLP	0.9879	6.33
	RF	0.9696	8.78
	DT	0.9690	8.81
	GBDT	0.9599	11.46
	LR	0.6576	37.75
TwrBsMx	MLP	0.9822	4.33
	RF	0.9445	11.20
	DT	0.9425	11.38
	GBDT	0.9360	13.58
	LR	0.7605	32.21

Overall, the MLP neural network achieved the best performance in the evaluation metrics across all three load channels, fully demonstrating the superiority of the neural network method in complex nonlinear load prediction tasks. Moreover, in a computing environment equipped with an Intel i7-12700K (12-core) CPU and 32 GB RAM, the average runtime of a 200 s OpenFAST co-simulation was approximately 20.6 s, whereas the trained MLP model required only 1.03 ms to predict the DEL for a single sample, indicating a substantial reduction in computational burden for fatigue load assessment.

Given that wind farm control parameter optimization involves numerous iterative evaluations, repeatedly calling high-fidelity simulations or performing real-time neural network predictions remains computationally demanding. To mitigate this issue, the trained MLP model was employed to perform offline DEL predictions under real wind speed samples for various power disturbance levels and control parameter combinations, and the results were organized into a multidimensional interpolation database. Specifically, the interpolation database was indexed by wind speed, power disturbance, virtual inertia coefficient, and droop coefficient, with the DEL values corresponding to the tower fore-aft bending moment, tower side-to-side bending moment, and rotor torque serving as response outputs. In the optimization phase, the DEL of each load channel was efficiently obtained through multidimensional interpolation using each turbine's real-time wind

speed, disturbance conditions, and control parameters, followed by the computation of the composite DEL index.

4. Wind Farm Control Parameter Optimization Strategy

4.1. Problem Description of Optimization

To resolve the trade-off between frequency support demands and fatigue load increase in wind farm primary frequency regulation, the virtual inertia and droop coefficients of each turbine were selected as decision variables, and a control parameter selective reduction optimization model incorporating system frequency security constraints was developed. This section formulates the problem from three perspectives—decision variables, objective function, and constraints—thereby providing a formal problem definition for the subsequent rolling optimization strategy.

For a wind farm consisting of 48 turbines, each turbine is assigned independent virtual inertia and droop coefficients, resulting in a total of 96 optimization variables, which can be expressed as:

$$X = [K_{df,1}, K_{pf,1}, \dots, K_{df,N}, K_{pf,N}] \quad (19)$$

where $N = 48$ denotes the number of wind turbines in the wind farm.

To simultaneously reduce overall fatigue load and improve load distribution uniformity among turbines, a composite objective function centered on total DEL and DEL variance is formulated. Specifically, total DEL characterizes the aggregate fatigue load of the wind farm, whereas DEL variance measures the dispersion of fatigue load distribution across turbines. Given the differences in units and numerical scales, the total DEL and DEL variance under the uniform control baseline are adopted as normalization references, denoted by DEL_{sum0} and DEL_{var0} , respectively. On this basis, the normalized composite objective function is formulated as follows:

$$F = \alpha \frac{DEL_{sum}}{DEL_{sum0}} + \beta \frac{DEL_{var}}{DEL_{var0}} \quad (20)$$

In this formulation, the objective function consists of two components: the normalized total DEL term and the normalized DEL variance term, representing the overall fatigue level and load distribution balance, respectively.

To emphasize the optimization principle of prioritizing total DEL with variance as a secondary objective under the condition of satisfying frequency constraints, the weighting coefficients are defined as:

$$\alpha = 1, \beta = 0.5 \quad (21)$$

The uniform control baseline state is obtained from the uniform control parameter search in the first stage of the subsequent rolling optimization method. In the uniform control baseline state, both normalized terms are equal to 1, resulting in an initial composite objective value of 1.50. It is important to emphasize that this composite index is mainly intended for monitoring the optimization process and for auxiliary comparison among candidate solutions under the same constraint level, rather than serving as the sole acceptance criterion. The acceptance of candidate solutions is determined based on a comprehensive evaluation of frequency constraint satisfaction and variations in total DEL and DEL variance.

The total DEL and DEL variance of the wind farm are defined as follows:

$$DEL_{sum} = \sum_{i=1}^N DEL_i \quad (22)$$

$$DEL_{var} = \frac{1}{N-1} \sum_{i=1}^N (DEL_i - \overline{DEL})^2 \quad (23)$$

where $\overline{DEL}$ is the mean value of the turbine DELs.

Furthermore, to uniformly characterize the fatigue level of a single turbine across multiple key load channels, the composite DEL index for the i -th turbine is defined as

$$DEL_i = DEL_{TwrBsMyt,i} + DEL_{TwrBsMxt,i} + DEL_{LSShftMxa,i} \quad (24)$$

This index is used to provide a unified representation of the DEL of three key load channels: tower fore–aft bending moment, tower side-to-side bending moment, and rotor torque. To avoid bias introduced by subjective weighting, equal weighting is applied to the three load channels.

In terms of constraints, considering that the primary prerequisite for wind farm participation in primary frequency regulation is to ensure system frequency security, system frequency constraints are treated as hard constraints in this study. Specifically, the constraints include the following categories.

First, to ensure that the system frequency does not fall below the safety limit after a disturbance, a frequency nadir constraint is imposed as follows:

$$f_{min} \geq f_{limit} \quad (25)$$

where f_{min} is the minimum value of the system frequency during the frequency response process, and f_{limit} is the lower bound constraint on the frequency nadir.

Secondly, to limit the rate of system frequency change after a disturbance, a rate of change of frequency (ROCOF) constraint is imposed as follows:

$$|RoCoF|_{max} \leq 1.0 \text{ Hz/s} \quad (26)$$

Furthermore, since the system frequency needs to return to the permissible operating range after primary frequency control, a recovery constraint is introduced, stipulating that the frequency should enter and remain within 49.5–50.5 Hz within 60 s following the disturbance.

In addition to frequency security constraints, boundary constraints on control parameters must also be satisfied during the optimization process. As the virtual inertia and droop coefficients are constrained by physical realizability and engineering adjustment limits, the parameter bounds are specified as follows:

$$\begin{aligned} K_{pf}^{min} &\leq K_{pf,i} \leq K_{pf}^{max} \\ K_{df}^{min} &\leq K_{df,i} \leq K_{df}^{max} \end{aligned} \quad (27)$$

The droop control coefficient K_{pf} is constrained within the range of 0–80 because excessively large droop gains may cause turbines operating at low wind speeds to provide excessive active power during frequency support, thereby causing rapid rotor speed reduction and increasing the risk of protection activation or shutdown. The virtual inertia coefficient K_{df} is limited to the range of 0–50, since excessively high inertia gains may lead to excessive sensitivity to the frequency rate of change, resulting in severe transient power fluctuations at the onset of the disturbance and thereby compromising power output smoothness and turbine stability.

In summary, the optimization problem can be formulated as follows: under system frequency constraints and control parameter bounds, the virtual inertia and droop coeffi-

coefficients of the turbines are selectively adjusted to minimize the overall fatigue load of the wind farm while improving the uniformity of load distribution among turbines.

4.2. Rolling Optimization Strategy

To address the high-dimensional, strongly constrained control parameter optimization problem with coupled frequency support and fatigue optimization, a two-stage rolling optimization method is proposed in this study. The method first searches for a feasible initial solution that satisfies the frequency constraints under a uniform control framework. On this basis, a differentiated control reduction optimization is carried out for individual turbines by incorporating DEL-based dynamic grouping, score-driven main-path search, and single-variable rollback strategies. The flow diagram of the rolling optimization method is given in Figure 4.

4.2.1. Phase I: Feasible Solution Initialization with Uniform Control

The first stage consists of a uniform control parameter search. In this stage, all turbines are assumed to operate with identical virtual inertia and droop coefficients, and feasible parameter combinations satisfying the frequency nadir constraint are identified within a uniform control framework; among them, the solution whose frequency nadir is closest to the constraint limit is selected as the baseline. This baseline state is not manually specified but is determined from the feasible solution search results. This uniform control baseline is then used as the initial operating point. Under this baseline, the system frequency response curve is obtained via simulation, and the DEL of each turbine as well as the total DEL of the wind farm are calculated using the previously established fast DEL evaluation model, thereby providing the initial evaluation basis for the second-stage rolling optimization.

4.2.2. Phase II: Rolling Optimization Based on DEL Dynamic Grouping Dynamic Grouping Based on DEL

In the second stage, to reduce optimization dimensionality and improve search stability, a DEL-based dynamic grouping mechanism is introduced. Specifically, turbines are sorted in descending order according to their DEL values and divided into high-, medium-, and low-DEL groups, each containing 16 turbines. To avoid the loss of adaptability caused by static grouping, a rolling update interval of three iterations is set, meaning that grouping is re-evaluated and updated based on the latest DEL results after every three optimization cycles. Through this iterative “Optimize–regroup–re-optimize” process, the original high-dimensional turbine-level optimization problem is converted into a lower-dimensional group-level differentiated optimization problem, significantly reducing search complexity and enhancing solution stability.

Scoring Metrics for Main Path Group Selection

After dynamic grouping, it is still necessary to determine which group’s control parameters should be prioritized for adjustment under the current grouping. If only average DEL values or a predefined sequence is used for searching, it cannot effectively reflect the fatigue benefit (total DEL reduction), frequency penalty (frequency nadir degradation), and load balancing variation (variance change) across different groups in the current state. To this end, a scoring metric for selecting the main path group is formulated at the beginning of each rolling optimization iteration, enabling comprehensive evaluation of candidate groups, and the group with the highest score is chosen as the main path group for the current iteration.

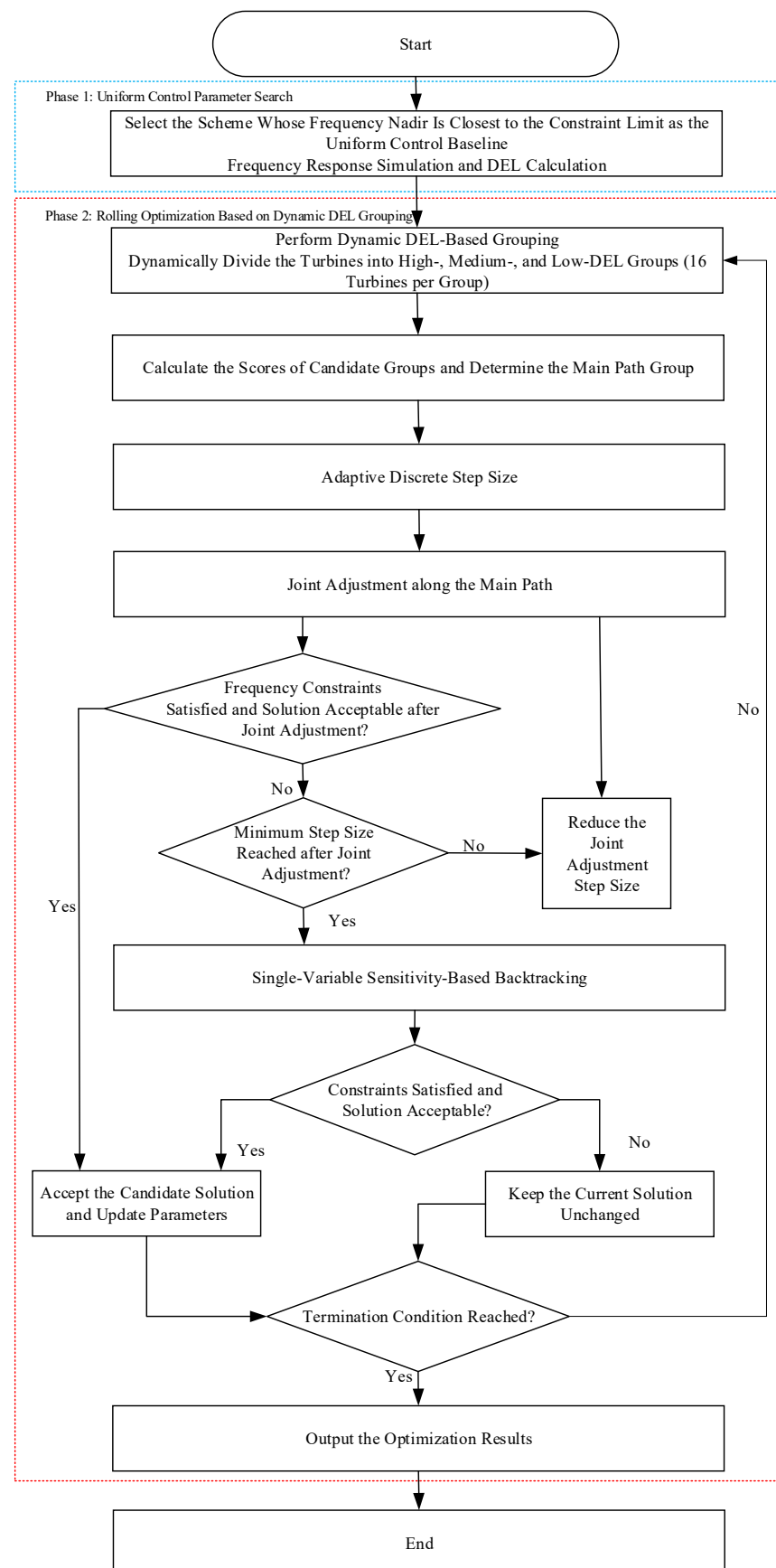


Figure 4. Rolling optimization process: the blue region denotes Phase I: Feasible Solution Initialization with Uniform Control, while the red region denotes Phase II: Rolling Optimization Based on DEL Dynamic Grouping.

Let the current grouping result be $G_1, G_2, \dots, G_M$, where M is the number of groups. At the current rolling iteration starting point (i.e., the solution after the previous optimization round), the total DEL of the wind farm, the system frequency nadir, and the DEL variance are denoted as $DEL_{sum,base}$, $f_{min,base}$, and $DEL_{var,base}$, respectively. For a candidate group G_g , a tentative control coefficient reduction is applied while keeping other groups unchanged, and the resulting wind farm total DEL, system frequency nadir, and DEL variance are computed, denoted as $DEL_{sum,g}$, $f_{min,g}$, and Var_g , respectively. Accordingly, the fatigue benefit of group g is defined as:

$$\Delta DEL_g = DEL_{sum,base} - DEL_{sum,g} \quad (28)$$

where a larger ΔDEL_g indicates a more significant fatigue reduction benefit resulting from the control coefficient reduction of this group.

Correspondingly, to characterize the impact of reducing control coefficients of this group on system frequency support capability, the frequency cost is defined as:

$$\Delta f_g = f_{min,base} - f_{min,g} \quad (29)$$

where a smaller Δf_g indicates a weaker adverse impact of the control coefficient reduction on the frequency nadir.

In addition, to reflect the impact of control coefficient reduction of candidate groups on load distribution balance among turbines, the relative change in DEL variance is defined as:

$$r_g = \frac{DEL_{var,g} - DEL_{var,base}}{DEL_{var,base}} \quad (30)$$

where $r_g \leq 0$ indicates that reducing the control coefficients of this group improves load distribution balance, while $r_g > 0$ indicates that it leads to further dispersion of DEL distribution.

Based on the above, the score of group g is defined as:

$$Score_g = \frac{\Delta DEL_g}{(\Delta f_g + \varepsilon) \times (1 + \lambda \times r_g)} \quad (31)$$

where ε is a small positive constant introduced to avoid division by zero, set to 0.0001; and λ is the weighting coefficient for load balance reward/penalty, set to 0.5.

A higher score implies that the candidate group achieves superior combined performance in fatigue reduction, frequency impact, and load balancing, making it more appropriate as the main path group in the current rolling optimization step. This scoring mechanism reflects the fatigue benefit, frequency cost, and improvement in load imbalance achieved by applying a one-step tentative reduction in control parameters to the candidate group relative to the current feasible solution. As these quantities evolve dynamically with the current DEL distribution and frequency margin, the scoring mechanism exhibits strong iteration dependence and state adaptiveness, thereby enabling more precise identification of candidate groups with higher optimization value under the current conditions.

Adaptive Discrete Step Size

Based on the above design, a hierarchical search mechanism is established for main-path group selection based on group scoring. After the main-path group scores are determined, a step-size parameter s is introduced to control the magnitude of parameter updates at each iteration. Since the control parameters are updated with a discretization interval of 0.5, the step size s is adjusted using the same interval to maintain consistency

between the search procedure and the parameter update mechanism. Furthermore, the frequency security margin is defined as the difference between the current frequency nadir and the lower frequency constraint limit, thereby characterizing the safety margin with respect to the constraint boundary. Additionally, the recent acceptance ratio is defined as the proportion of accepted candidate solutions over a predefined number of recent iterations, reflecting the stability of the current search process. Based on these two indicators, a conservative discrete step-size adjustment strategy is adopted, in which step reduction is prioritized and step enlargement is used only sparingly: when the frequency security margin is small or the acceptance ratio is low, the step size is reduced to limit parameter perturbations and improve feasibility. The step size is increased moderately only when the frequency security margin is adequate and the search process is stable, so as to improve search efficiency. Therefore, this step-size strategy is not intended to aggressively broaden the search space, but rather to improve the robustness and feasibility of local search in the vicinity of the frequency constraint boundary.

Hierarchical Search Mechanism

In the search procedure, this paper adopts a hierarchical search mechanism termed “Main-path joint adjustment–single-variable sensitivity rollback”. The first step consists of main-path joint adjustment, in which the units within the current main-path group synchronously reduce the virtual inertia and droop coefficients to prioritize the exploration of the fatigue-reduction potential arising from their coordinated variation. For candidate solutions obtained via joint adjustment, it is first verified whether the frequency constraints are satisfied; if so, their optimization performance is further evaluated. A candidate solution is accepted if the total DEL decreases, or if the total DEL increases by no more than 1% while the composite metric is no worse than that of the current solution. If the candidate solution obtained from joint adjustment is not accepted under the current step size, the step size is preferentially reduced, and the search continues. The process proceeds to the single-variable adjustment stage only when no acceptable solution can be found after the joint-adjustment step size has been reduced to its minimum value.

The second step is single-variable adjustment, in which two paths are sequentially explored: (i) reducing only the virtual inertia coefficient of the current high-scoring group while keeping the droop coefficient unchanged; (ii) reducing only the droop coefficient while keeping the virtual inertia coefficient unchanged. The purpose of introducing single-variable adjustment is to further identify the distinct impacts of virtual inertia and droop coefficients on frequency feasibility and fatigue performance under the current operating condition when joint adjustment is infeasible or insufficiently beneficial, thereby finding more robust alternative update strategies. Candidate solutions obtained from single-variable adjustment are evaluated according to the same criteria as those used for joint adjustment.

Termination Criterion

To prevent ineffective iterations or excessive search during the rolling optimization process, multiple criteria are adopted to terminate the procedure, including: (1) reaching the maximum number of iterations; (2) no acceptable update for three consecutive iterations after the step size has been reduced to its minimum; (3) the variance of DEL decreases to below 5% of its initial value; and (4) the composite objective function enters a plateau region with limited improvement over several consecutive iterations. Among these, the first two criteria are mainly used to prevent ineffective search and excessive iterations, while the latter two are designed to identify conditions where load balancing is essentially achieved or the objective function has entered a plateau phase, thereby improving computational efficiency while ensuring optimization performance.

5. Simulation System and Results Analysis

To verify the effectiveness of the proposed optimization method for virtual inertia-based frequency regulation control parameters of wind farms based on fatigue load prediction, a simulation model of wind farm participation in primary frequency regulation is established on the MATLAB/Simulink platform. The study system is the IEEE 9-bus system integrated with a 240 MW wind farm, where the wind farm consists of 48 NREL 5 MW wind turbines, each equipped with virtual inertia control and droop control loops. Measured wind speed data from the LY wind farm are used to characterize turbine operating conditions, with a steady-state output of 138.4 MW; the wind speed profile is shown in Figure 5.

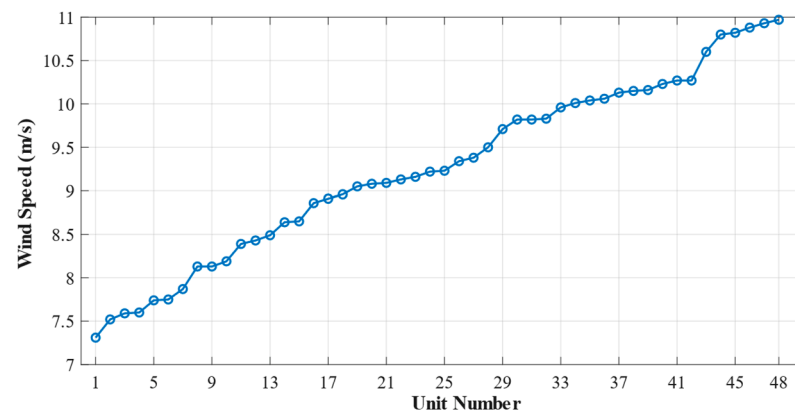


Figure 5. Wind speeds of each wind turbine.

Simulation results indicate that when the wind farm does not participate in frequency regulation, the system frequency nadir is only 49.65 Hz, which fails to meet the frequency security requirements. In this study, the system frequency nadir is constrained to be no lower than 49.8 Hz, serving as a frequency security constraint for the optimization of wind farm fatigue loads. Figure 6 illustrates the system frequency response without wind farm participation in frequency regulation.

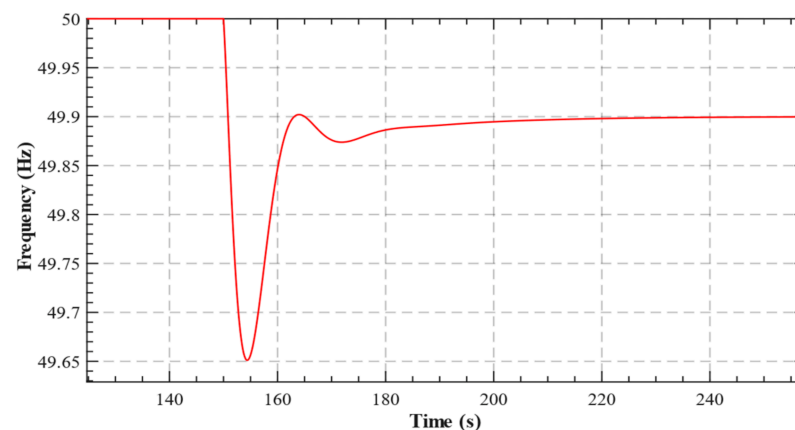


Figure 6. Frequency curve without wind turbine participation in frequency regulation.

5.1. Baseline with Uniform Control Parameters

In this paper, the uniform control parameter scheme is adopted as the baseline case. Through a search under uniform control parameters, it is found that when $K_{pf} = 30$ and $K_{df} = 10$, the system frequency response satisfies the frequency constraint, with a frequency nadir of 49.8025 Hz, which is above the lower limit of 49.8 Hz. The control coefficients obtained from this uniform scheme are used as the baseline operating point

for subsequent rolling optimization, enabling further investigation of the effects of differentiated derating control on both the total fatigue load and its distribution uniformity under frequency security constraints. Under the baseline condition, the total DEL of the wind farm is 81,191.08, with an average DEL of 1691.48, a standard deviation of 111.06, a variance of 12,334.82. This indicates that although the uniform control scheme satisfies the frequency support requirements, there are significant differences in fatigue load distribution among turbines, suggesting considerable room for further optimization. Subsequently, DEL dynamic grouping-based rolling optimization is performed based on this scheme.

5.2. Analysis of the Rolling Optimization Process

Figure 7 presents a comprehensive analysis of the optimization process, including the evolution of total DEL, DEL variance, composite objective function, and frequency nadir.

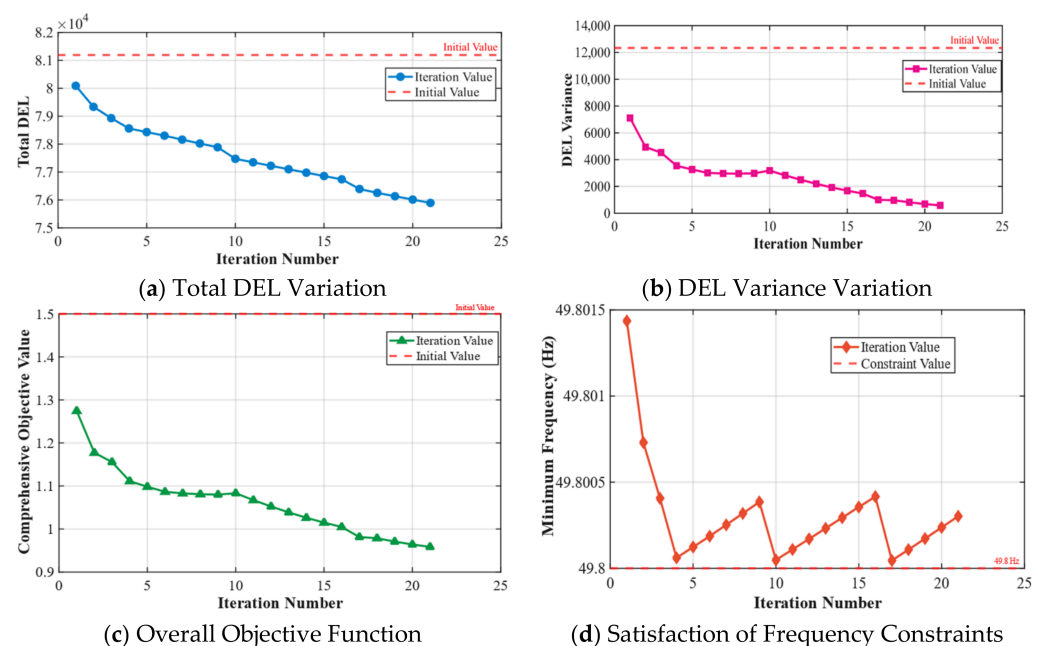


Figure 7. Optimization process.

First, as shown in Figure 7a, the total DEL convergence curve exhibits an overall decreasing trend as the iterations proceed. The rapid decrease in the initial iterations indicates that the grouping evaluation metric, constructed based on fatigue benefit, frequency cost, and load balancing variation, can effectively identify candidate groups with greater optimization potential under the current conditions. As the optimization proceeds, the rate of decrease in total DEL gradually diminishes, indicating that the available space for further fatigue optimization becomes increasingly limited under tightening frequency constraints. Ultimately, the total DEL decreases to 75,893.65 after the 21st iteration, and the termination criteria are met before the maximum number of iterations is reached.

Secondly, as shown in Figure 7b, the DEL variance decreases from the initial 12,334.82 to 583.53, indicating a significant improvement in the uniformity of fatigue load distribution among turbines. Particularly in the early iterations, the variance decreases rapidly, indicating that the proposed method can not only reduce the overall fatigue level but also narrow the load differences among turbines. As the optimization proceeds, the dynamic grouping mechanism continuously updates candidate groups based on the latest DEL distribution, enabling subsequent search to target the current fatigue distribution state. In fact, the algorithm triggers the highly balanced termination criterion when the DEL variance decreases to approximately 4.7% of its initial value (i.e., below 5% of the initial variance).

Next, as shown in Figure 7c, the composite objective function decreases continuously from the initial value of 1.50 to 0.96, exhibiting an overall downward trend consistent with the variations in total DEL and DEL variance. It is noteworthy that around iterations 6–10, the decrease in the objective function becomes gradual and even exhibits slight fluctuations. This phenomenon indicates that, as the optimization progresses, the acceptability of candidate solutions gradually decreases, and the search process may increasingly rely on more conservative local adjustment paths, thereby exhibiting a staged plateau characteristic.

Finally, Figure 7d shows that the frequency nadir is maintained above the 49.8 Hz constraint throughout the entire optimization process, demonstrating that the proposed method consistently enforces the frequency security requirement. Meanwhile, the frequency nadir fluctuates near the constraint boundary, suggesting that the optimization process operates in the vicinity of the frequency constraint limit, aiming to maximize fatigue reduction potential under guaranteed frequency security.

5.3. Analysis of Optimization Results

After applying the proposed optimization method, the fatigue load indicators of the wind farm are significantly improved. Table 6 presents a comparison of various parameters between the optimized results and the baseline scheme. Compared with the uniform control baseline scheme, the optimized total DEL of the wind farm decreases from 81,191.08 to 75,893.6, with a reduction of 5297.6, corresponding to a decrease of 6.5%. The average DEL declines from 1691.48 to 1581.11, corresponding to a 6.5% decrease. The DEL standard deviation is reduced from 111.06 to 24.16, a decrease of 78.25. The DEL variance drops from 12,334.82 to 583.54, corresponding to a 95.27% reduction. The normalized composite objective value is reduced from 1.50 to 0.96, achieving a 36.12% improvement. The above results demonstrate that, while satisfying frequency security constraints, the proposed method effectively reduces the overall fatigue level of the wind farm and significantly improves the imbalance in fatigue load distribution among turbines.

Table 6. Comparison of parameters between optimized results and baseline scheme.

Method	Total DEL	Average DEL	Standard Deviation	Variance	Comprehensive Objective
Baseline	81,191.08	1691.48	111.06	12,334.82	1.50
Optimized	75,893.65	1581.11	24.16	583.53	0.96

Figure 8 presents a comparison of the aggregated DEL of 48 turbines before and after optimization. Under the baseline scheme, the DEL distribution is clearly uneven across turbines, as evidenced by the pronounced fluctuations in the DEL curve. This suggests that although uniform control parameters maintain control consistency, they fail to account for inter-turbine coordination under heterogeneous wind conditions, thereby causing some turbines to bear relatively high fatigue loads. After optimization, the DEL curve becomes markedly flatter, and the DEL values of most turbines are concentrated within approximately 1540–1650. Relative to the baseline scheme, turbines with originally high DEL values exhibit a substantial reduction in fatigue load, particularly those located at the beginning and end of the curve, whereas turbines with originally low DEL values show little change. These results indicate that, under the current frequency constraints, the proposed method improves the internal DEL distribution of the wind farm mainly by reducing the fatigue load levels of turbines with high and moderate fatigue.

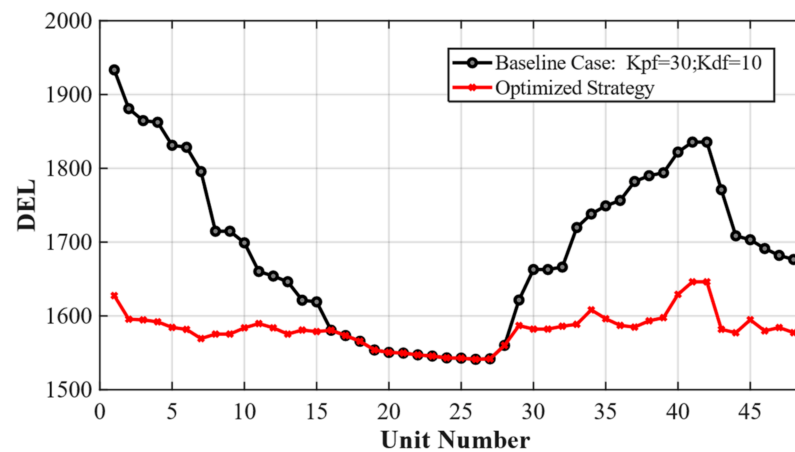


Figure 8. Comparison of DEL curves of each wind turbine between the optimized result and the baseline scheme.

Figure 9 shows the comparison of the distribution of K_{pf} and K_{df} parameters for each turbine before and after optimization. After optimization, the control parameters of different turbines exhibit clear differentiation.

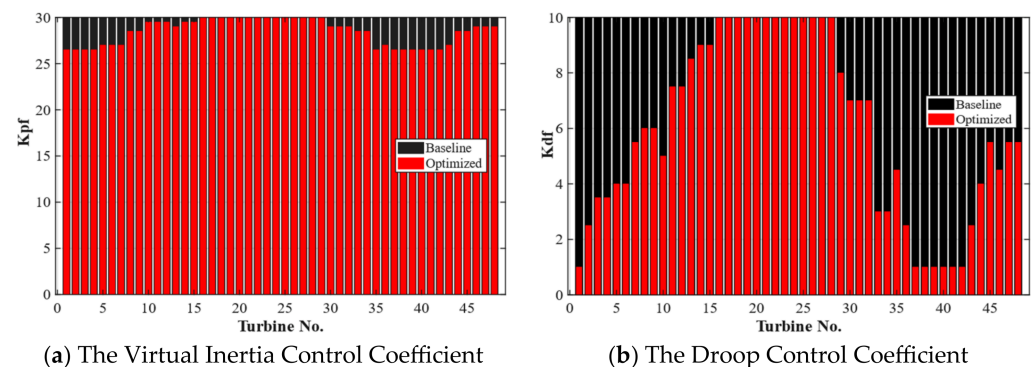


Figure 9. Comparison of the control coefficients of individual wind turbines between the optimized result and the baseline scheme: (a) comparison of the virtual inertia control coefficient; (b) comparison of the droop control coefficient.

From the K_{df} distribution in Figure 9a, it can be observed that the virtual inertia coefficient exhibits substantial variation after optimization, ranging from 0 to 10 with an average value of 6. The K_{df} values of turbines corresponding to high- and medium-DEL regions are significantly reduced, in some cases to as low as 1, while turbines with low DEL retain relatively high K_{df} values. This indicates that the virtual inertia coefficient serves as the dominant tuning variable in the fatigue optimization process. Lowering the K_{df} of high-fatigue turbines effectively attenuates their rapid power response to rate-of-change-of-frequency signals, thereby limiting the increase in additional fatigue loads. In contrast, the variation of K_{pf} in Figure 9b is relatively small, with an overall range of 26.5–30 and an average value of 28.5 after optimization, and most turbines remain close to their initial values. This phenomenon is consistent with the algorithm design and simulation results: since the droop coefficient K_{pf} is directly related to frequency deviation, its reduction more directly weakens the sustained frequency support capability of turbines, thereby making the frequency nadir more susceptible to decline.

Therefore, when joint adjustment becomes infeasible, the algorithm mainly realizes fatigue optimization through reductions in K_{df} , while keeping the adjustment of K_{pf} relatively conservative. It is also noteworthy that the optimized parameters are adjusted downward relative to the uniform baseline values rather than shifted toward the predefined

upper bounds. This behavior is consistent with the selective derating control formulation adopted in this study and suggests that, under frequency constraints, the proposed method can effectively coordinate frequency-support performance with fatigue load reduction.

Figure 10 shows the comparison of power responses between the optimized results and the uniform control scheme. To illustrate the impact of differentiated parameter adjustment on the frequency regulation burden of individual turbines, the turbine with the largest DEL reduction is selected as a representative example. Figure 10a presents the comparison of power responses of Turbine 1, which exhibits the largest DEL change, before and after optimization. Under the baseline scheme, the aggregated DEL of this turbine is 1933.4, which decreases to 1627.5 after optimization, corresponding to a reduction of 15.8%, the largest among all turbines. Correspondingly, its control parameters are adjusted from $K_{pf} = 30$ and $K_{df} = 10$, to $K_{pf} = 26.5$ and $K_{df} = 1.0$ indicating that both virtual inertia support and droop support capabilities are significantly weakened, with a particularly pronounced reduction in K_{df} . This indicates that the optimized turbine undertakes significantly less active power regulation throughout the frequency regulation process. Since tower bending moment and main shaft torque fatigue are closely related to rapid power variations, the significant reduction in DEL for this turbine is reasonable. Figure 10b shows the comparison of the total power response of the wind farm before and after optimization. It can be observed that the total power response curves under the baseline and optimized schemes almost overlap, with only slight differences at the peak and recovery stages. This indicates that the optimized scheme achieves improvements in fatigue metrics while maintaining a nearly identical overall power response profile.

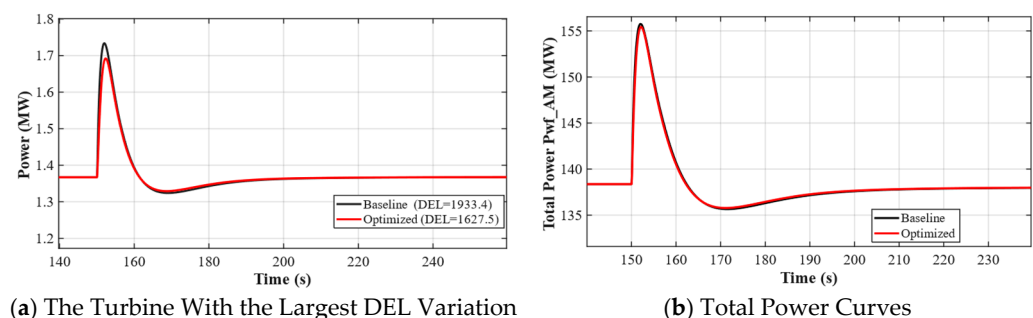


Figure 10. Comparison of power curves between the optimized result and the baseline scheme: (a) comparison of the power curves of the turbine with the largest DEL variation; (b) comparison of the total power curves.

Figure 11 presents a comparison of frequency and rate of change of frequency (ROCOF). Figure 11a shows a comparison of the system frequency response curves between the baseline and optimized schemes. It can be observed that both schemes satisfy the frequency constraints defined in this study. Following the disturbance, the two schemes exhibit similar frequency trajectories, with a rapid decline followed by gradual recovery under wind farm frequency support. The frequency nadir is 49.8025 Hz for the baseline scheme and 49.8003 Hz for the optimized scheme, both remaining above the 49.8 Hz limit. Compared with the baseline scheme, the optimized scheme shows a slightly lower nadir, with a reduction of approximately 0.0022 Hz, indicating that the support capability of the main-path turbines is moderately weakened during fatigue optimization and that the nadir moves closer to the constraint boundary. Figure 11b further shows that the ROCOF curves of the two schemes follow similar overall trends. The maximum ROCOF is 0.1045 Hz/s in the baseline case and 0.1067 Hz/s after optimization, suggesting a slightly faster initial frequency decline after the disturbance. This is consistent with the marginal reduction in the frequency nadir. Nevertheless, the overall difference in ROCOF remains small, indicating

that the dynamic frequency performance is not significantly degraded. Therefore, although the optimization slightly weakens the initial inertial response, its impact remains within acceptable limits as all frequency constraints are satisfied. These results confirm that the proposed method can coordinate frequency-support performance and fatigue load reduction under frequency constraints.

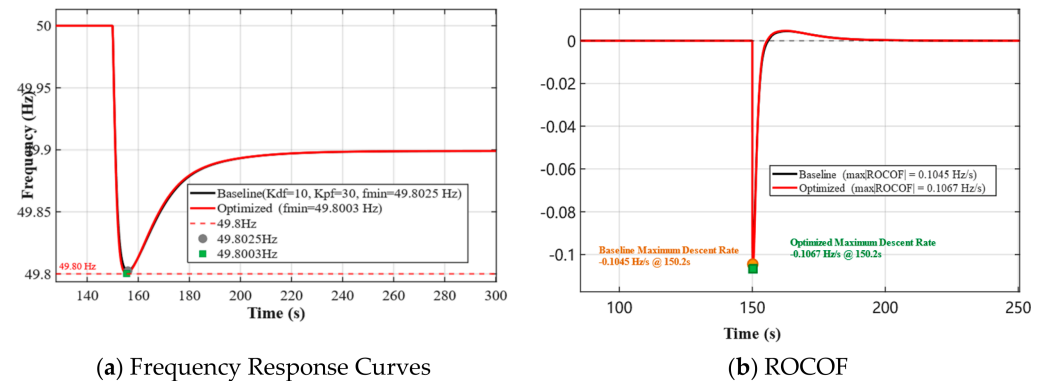


Figure 11. Comparison of frequency responses between the baseline scheme and the optimized scheme: (a) comparison of system frequency response curves; (b) comparison of the system rate of change of frequency (ROCOF).

5.4. Strategy Table Based on the Wind Speed Combination Model

The above simulation results demonstrate the feasibility of the proposed method under real wind speed conditions with medium disturbance. However, considering the high computational cost of the simulation model used in this study, a single complete rolling optimization process is time-consuming and cannot directly meet real-time on-line solution requirements in practical applications. However, wind farm participation in primary frequency regulation requires rapid determination of control parameters in response to disturbance levels; thus, re-running the full optimization procedure after each disturbance is not feasible for practical deployment. To enable practical implementation of wind farm primary frequency regulation control parameters, a strategy table of control parameters under various operating conditions is constructed. The strategy table is indexed by wind farm steady-state output power and power deficit level, with 135 MW and 140 MW selected as representative operating conditions, and the corresponding turbine wind speed distributions determined using a wind speed combination model [35]. Using a steady-state output power of 138.4 MW under real wind speed conditions as an example, a comparison between actual wind speeds and those obtained from the wind speed combination model is provided. Using a steady-state output power of 138.4 MW under real wind speed conditions as an example, Figure 12 illustrates the comparison between actual wind speeds and those generated by the wind speed combination model.

In this study, a medium disturbance scenario with a 35 MW power deficit is taken as an example, and the frequency nadir target is set to 49.75 Hz. Accordingly, two disturbance levels of power deficit, 0.05 and 0.06, are selected in the strategy table. For the 0.06 condition, the frequency nadir constraint of not lower than 49.75 Hz is maintained for parameter selection. For the 0.05 condition, considering that the control parameters should maintain adaptability to slightly larger disturbances close to 0.0506, an additional verification under the 0.055 condition is performed. Accordingly, the target frequency nadir for the 0.05 condition is set to 49.775 Hz. In this way, a control parameter strategy table under different steady-state output powers and disturbance levels can be constructed. Table 7 presents the summary of uniform control baseline schemes under the strategy table.

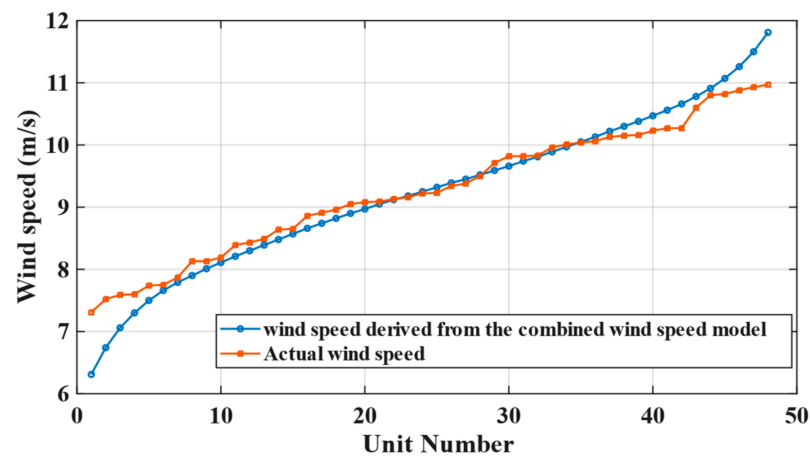


Figure 12. Comparison of actual wind speeds and wind speeds derived from the wind speed combination model.

Table 7. Summary table of uniform control baseline schemes under the strategy table.

Power Disturbance (p.u.)	Steady-State Output Power (MW)	Original Frequency Nadir (Hz)	Minimum Frequency Requirement (Hz)	Kdf Under Uniform Control	Kpf Under Uniform Control	Frequency Nadir Under the Baseline Scheme (Hz)
0.05	135	49.568	49.775	15	23	49.7754
	140	49.568	49.775	14	23	49.7755
0.06	135	49.655	49.75	30	30	49.7508
	140	49.655	49.75	30	30	49.7505

Figures 13–16 present the optimized control parameter results under the strategy table operating conditions.

Under the actual condition of a power deficit $\Delta P_d = 35/692$ p.u. and a steady-state output power of 138.4 MW, the control parameters are determined according to the strategy corresponding to $\Delta P_d = 0.05$ p.u. and $p = 140$ MW based on the nearest-level principle. The corresponding frequency response curve is shown in Figure 17. The results indicate that, with the optimized control, the frequency nadir reaches 49.7728 Hz, exceeding 49.75 Hz, and the frequency response meets the required criteria.

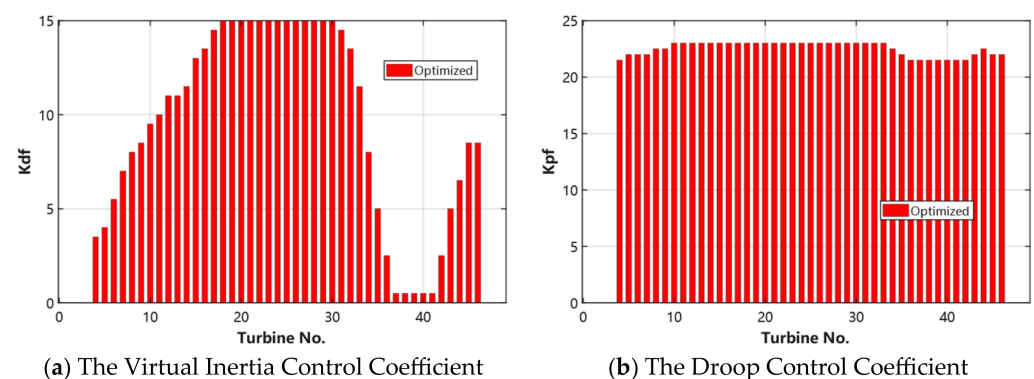


Figure 13. Optimized control parameters under $\Delta P_d = 0.05$ p.u., $p = 135$ MW.

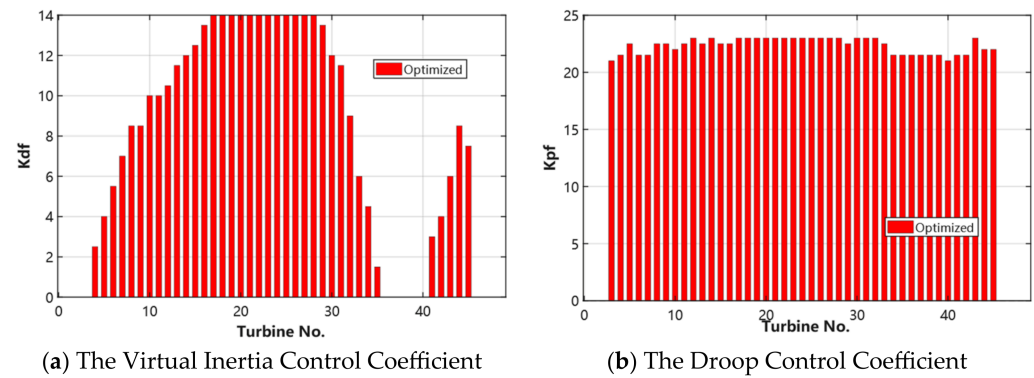


Figure 14. Optimized control parameters under $\Delta P_d = 0.05$ p.u., $p = 140$ MW.

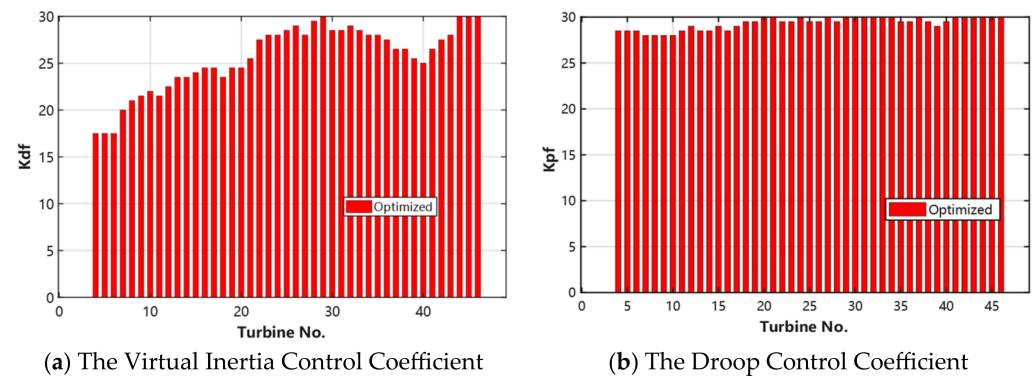


Figure 15. Optimized control parameters under $\Delta P_d = 0.06$ p.u., $p = 135$ MW.

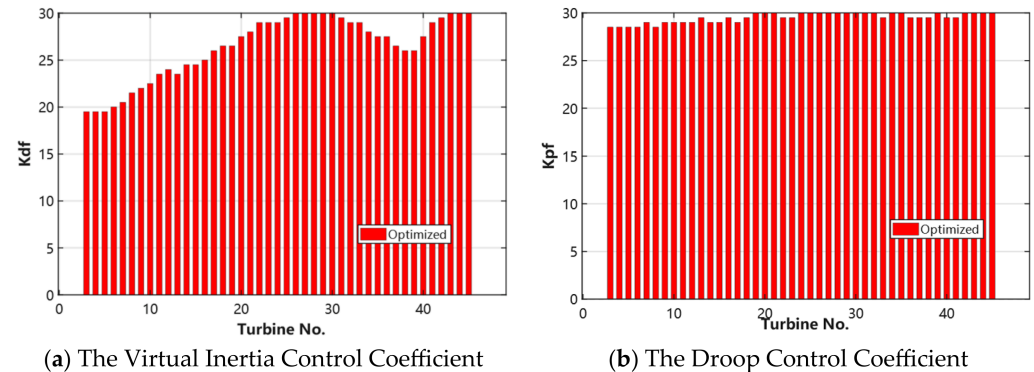


Figure 16. Optimized control parameters under $\Delta P_d = 0.06$ p.u., $p = 140$ MW.

Table 8 presents the optimization results obtained using the strategy table. Compared with the uniform control baseline parameters in the strategy table, the optimized parameters reduce the total DEL of the wind farm from 70,623.41 to 64,905.38, corresponding to a reduction of 8.10%. Meanwhile, the variance of the aggregated DEL across the 48 wind turbines decreases from 58,061.86 to 38,914.78, with a reduction of 32.98%. These results indicate that the strategy table scheme reduces the overall fatigue load level while improving the balance of load distribution among turbines.

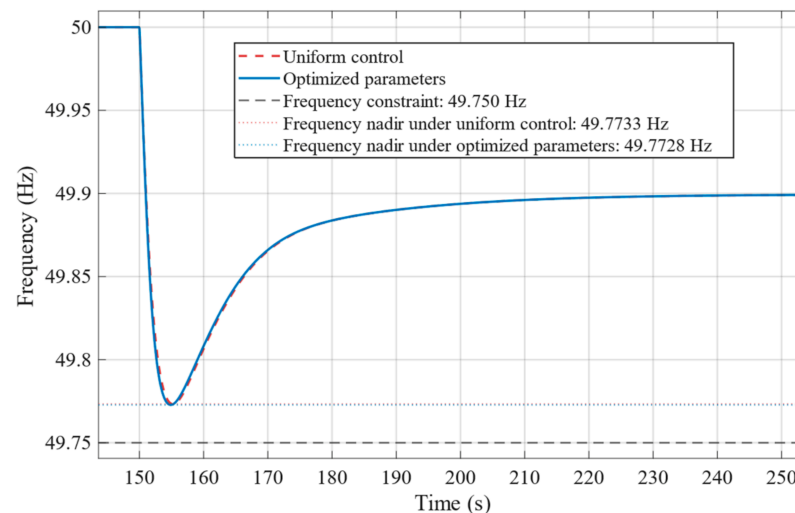


Figure 17. Frequency curve.

Table 8. The optimization results obtained using the strategy table.

Method	Total DEL	Average DEL	Standard Deviation	Variance
Baseline	70,623.41	1471.32	240.96	58,061.86
Optimized	64,905.38	1352.20	197.27	38,914.78

6. Discussion

There exists an inherent trade-off between wind turbine participation in primary frequency regulation and structural fatigue mitigation. The preceding mechanism analysis has shown that virtual inertia and droop control increase the dynamic variation in active power, thereby intensifying fluctuations in drivetrain torque and tower bending moments, which ultimately result in an increase in DEL. Therefore, in primary frequency regulation scenarios, reducing the overall fatigue load of a wind farm inherently requires a certain degree of reduction in the control coefficients, which in turn weakens the frequency-support capability of the wind farm. This implies that the problem addressed in this paper is not to eliminate the conflict between frequency-regulation performance and fatigue mitigation, but rather to identify the maximum feasible reduction margin of the control parameters under given frequency-security constraints so as to obtain fatigue-reduction benefits.

Based on the above analysis, the core of the proposed method is to identify candidate groups with high adjustment priority under the current operating conditions. During the rolling optimization process, tentative reductions in the control parameters are first imposed on each candidate group, after which a composite score is evaluated according to fatigue benefit, frequency cost, and improvement in DEL distribution. Groups with higher scores are then prioritized for subsequent exploration. Therefore, the proposed method is essentially a frequency-constrained selective derating optimization strategy, rather than a scheme that uniformly reduces control parameters across all groups. Notably, the main-path group is determined according to the composite adjustment score obtained after tentative parameter reduction rather than according to the initial DEL level alone. Together with the simulation results presented above, this confirms that the proposed scoring mechanism can effectively identify candidate groups with high adjustment potential under the prevailing constraints.

From a methodological standpoint, different control parameters play distinct roles in the coupling between frequency support and structural fatigue. Virtual inertia control

primarily affects the fast power response at the onset of a disturbance, and is therefore more likely to induce transient fluctuations in drivetrain torque and tower loads, making it more sensitive to DEL variations. By contrast, droop control mainly determines the support level during the sustained frequency-deviation stage and has a more direct influence on the system frequency nadir. Consequently, under frequency constraints, the optimization tends to prioritize adjustment of the virtual inertia coefficient for fatigue reduction, while treating the droop coefficient more conservatively. This distinction further indicates that, in the fatigue optimization of primary frequency regulation, different control parameters exhibit different adjustment margins and different sensitivities to frequency-security constraints.

In addition, the optimization results are strongly constrained by the system frequency requirements. When the frequency nadir of the uniform-control baseline is already close to the constraint boundary, the available frequency margin becomes limited, and the scope for further reduction in the control parameters is correspondingly restricted. Under such conditions, the proposed method exhibits a conservative derating behavior, in which the available adjustment degrees of freedom with greater optimization value are preferentially exploited to reduce fatigue while preventing significant deterioration in frequency-support capability.

7. Conclusions

This study focuses on the coordination between frequency support requirements and structural fatigue damage during wind farm participation in primary frequency control and proposes an integrated control-parameter optimization method based on data-driven fatigue load assessment. First, a DEL sample database under frequency regulation conditions is developed via co-simulation using OpenFAST and MATLAB/Simulink, and a data-driven mapping model is established to relate power disturbance, wind speed, virtual inertia coefficient, and droop coefficient to the DEL of key fatigue load channels. On this basis, a control parameter optimization model is developed to simultaneously reduce the overall fatigue load and improve the uniformity of DEL distribution among turbines. To tackle the high-dimensional control parameter space and stringent frequency constraints in large-scale wind farms, a main-path rolling optimization strategy based on DEL dynamic grouping and composite evaluation is proposed, enabling differentiated reduction in control parameters across turbines.

Simulation results show that although the uniform control scheme satisfies the system frequency nadir constraint, significant differences remain in the fatigue load distribution among turbines, indicating further optimization potential. After applying the proposed method, while satisfying the system frequency security constraints, both the total DEL and DEL variance of the wind farm are significantly reduced, indicating an effective reduction in the overall fatigue level and an improvement in DEL distribution uniformity. The optimization results are manifested in differentiated adjustments of the control parameters, primarily reflected in the reduced fatigue levels of turbines located in high- and medium-DEL regions in this case study. These results indicate that, in wind farm primary frequency regulation with fatigue effects taken into account, control parameter optimization should adopt differentiated reductions according to turbine fatigue states under frequency security constraints, thereby achieving coordinated optimization between fatigue mitigation and frequency support. Furthermore, considering that the complete rolling optimization process is time-consuming and currently difficult to meet real-time online solution requirements, a control strategy table for medium disturbance levels is constructed based on offline optimization results. The strategy table enables rapid matching and deployment of corresponding control parameter configurations after a disturbance occurs, thereby improving deployment efficiency while preserving optimization performance. These results indicate

that the proposed method not only provides a feasible approach for optimizing integrated virtual inertia parameters in wind farms while balancing frequency security and structural reliability but also lays the foundation for fast parameter deployment and online control implementation in practical engineering applications.

This study is presently based on a single disturbance scenario and a predefined wind speed distribution and does not yet account for complex operational factors such as stochastic wind variations, wake interactions, and multi-period consecutive disturbances. Future research may focus on robust optimization under multiple scenarios to improve the engineering applicability of the proposed method.

Author Contributions: Conceptualization, Y.J.; writing—original draft preparation, Y.J.; formal analysis, M.S.; methodology, M.S.; validation, M.S.; investigation, M.S.; writing—review and editing, M.S. and X.S. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by Smart Grid-National Science and Technology Major Project, Project No. 2024ZD0801400 and Science and technology projects of State Grid Corporation of China, Project No. 52272224000V.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The raw data supporting the conclusions of this article will be made available by the authors upon request.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Lim, S.; Kim, T.; Lee, K.Y.; Choi, D.; Park, J.-W. Impact of High Penetration of Renewables on Power System Frequency Response: A Review and Verification. *Renew. Sustain. Energy Rev.* **2025**, *217*, 115728. [\[CrossRef\]](#)
2. Zhang, G.; Ren, J.; Zeng, Y.; Liu, F.; Wang, S.; Jia, H. Security Assessment Method for Inertia and Frequency Stability of High Proportional Renewable Energy System. *Int. J. Electr. Power Energy Syst.* **2023**, *153*, 109309. [\[CrossRef\]](#)
3. Radaelli, L.; Martinez, S. Frequency Stability Analysis of a Low Inertia Power System with Interactions among Power Electronics Interfaced Generators with Frequency Response Capabilities. *Appl. Sci.* **2022**, *12*, 11126. [\[CrossRef\]](#)
4. He, C.; Geng, H.; Rajashekara, K.; Chandra, A. Analysis and Control of Frequency Stability in Low-Inertia Power Systems: A Review. *IEEE/CAA J. Autom. Sin.* **2024**, *11*, 2363–2383. [\[CrossRef\]](#)
5. El-Bahay, M.H.; Lotfy, M.E.; El-Hameed, M.A. Computational Methods to Mitigate the Effect of High Penetration of Renewable Energy Sources on Power System Frequency Regulation: A Comprehensive Review. *Arch. Comput. Methods Eng.* **2023**, *30*, 703–726. [\[CrossRef\]](#)
6. Besheer, A.H.; Liu, X.; Eissa, S.F.; Rabah, M.; Mahgoub, A.; Rashad, H. Overview on Fast Primary Frequency Adjustment Technology for Wind Power Future Low Inertia Systems. *Alex. Eng. J.* **2023**, *78*, 318–338. [\[CrossRef\]](#)
7. Li, L.; Zhu, D.; Zou, X.; Hu, J.; Kang, Y.; Guerrero, J.M. Review of Frequency Regulation Requirements for Wind Power Plants in International Grid Codes. *Renew. Sustain. Energy Rev.* **2023**, *187*, 113731. [\[CrossRef\]](#)
8. Asad, M.; Sánchez-Fernández, J.Á. Frequency Regulation Provided by Doubly Fed Induction Generator Based Variable-Speed Wind Turbines Using Inertial Emulation and Droop Control in Hybrid Wind–Diesel Power Systems. *Appl. Sci.* **2025**, *15*, 5633. [\[CrossRef\]](#)
9. Chen, W.; Zheng, T.; Nian, H.; Yang, D.; Yang, W.; Geng, H. Multi-Objective Adaptive Inertia and Droop Control Method of Wind Turbine Generators. *IEEE Trans. Ind. Appl.* **2023**, *59*, 7789–7799. [\[CrossRef\]](#)
10. Bao, W.; Ding, L.; Kang, Y.C.; Sun, L. Closed-Loop Synthetic Inertia Control for Wind Turbine Generators in Association with Slightly Over-Speeded Deloading Operation. *IEEE Trans. Power Syst.* **2023**, *38*, 5022–5032. [\[CrossRef\]](#)
11. Wang, Y.; Guo, Y.; Zhang, D.; Liu, H.; Song, R. Analysis and Mitigation of the Drive Train Fatigue Load for Wind Turbine with Inertial Control. *Int. J. Electr. Power Energy Syst.* **2022**, *136*, 107698. [\[CrossRef\]](#)
12. Wang, Y.; Guo, Y.; Chen, Y.; Xu, W. Analysis of Frequency Regulation Capability and Fatigue Loads of Wind Turbine Based on Over-Speed Control. *Electronics* **2023**, *12*, 2009. [\[CrossRef\]](#)
13. Wang, Y.; Guo, Y.; Xu, W. Impact Mechanism of Frequency Response on Wind Turbine Fatigue Load. *J. Renew. Sustain. Energy* **2023**, *15*, 033304. [\[CrossRef\]](#)

14. Zhang, X.; He, W.; Hu, J. Impact of Inertia Control of DFIG-Based WT on Torsional Vibration in Drivetrain. *IEEE Trans. Sustain. Energy* **2020**, *11*, 2525–2534. [\[CrossRef\]](#)
15. Zhang, B.; Wang, Y.; Guo, Y.; Xu, W.; Jiang, X.; Ge, C. A Combined Wind-Storage Primary Frequency Regulation Method Considering Low-Speed Shaft Fatigue Loads in Wind Turbine. *Electr. Power Syst. Res.* **2024**, *235*, 110891. [\[CrossRef\]](#)
16. Jin, X.; Tan, W.; Zou, Y.; Wang, Z. Active Disturbance Rejection Control for Wind Turbine Fatigue Load. *Energies* **2022**, *15*, 6178. [\[CrossRef\]](#)
17. Pei, M.; Wang, Q.; Ye, L.; Luo, Y.; Sha, L.; Zhang, Z.; Song, X. Hierarchical Control Strategy of Wind-Storage Frequency Support for SOC Recovery Optimization and Arbitrage Revenue. *Appl. Energy* **2024**, *365*, 123229. [\[CrossRef\]](#)
18. Wang, D.; Guo, Y.; Huang, Y.; Zhou, K.; Wang, S. Torsional Vibration Analysis of Virtual-Synchronous-Controlled DFIG-Based Wind Turbines. *IEEE Trans. Sustain. Energy* **2025**, *16*, 2044–2057. [\[CrossRef\]](#)
19. Haselibozechaloe, D.; Correia, J.; Mendes, P.; De Jesus, A.; Berto, F. A Review of Fatigue Damage Assessment in Offshore Wind Turbine Support Structure. *Int. J. Fatigue* **2022**, *164*, 107145. [\[CrossRef\]](#)
20. Haghi, R.; Crawford, C. Data-Driven Surrogate Model for Wind Turbine Damage Equivalent Load. *Wind Energy Sci.* **2024**, *9*, 2039–2062. [\[CrossRef\]](#)
21. Haghi, R.; Stagg, C.; Crawford, C. Wind Turbine Damage Equivalent Load Assessment Using Gaussian Process Regression Combining Measurement and Synthetic Data. *Energies* **2024**, *17*, 346. [\[CrossRef\]](#)
22. Guo, Y.; Li, L.; Gui, J.; Hu, S. Neural Network Approaches for Real-Time Fatigue Life Estimation by Surrogating the Rainflow Counting Method. *Int. J. Fatigue* **2025**, *197*, 108941. [\[CrossRef\]](#)
23. Lyu, X.; Liu, T. Intelligent Learning Based Active Power Regulation of Wind Turbines Considering Fatigue Reduction. *IEEE Trans. Ind. Inform.* **2022**, *18*, 405–414. [\[CrossRef\]](#)
24. Fu, L.; Liu, J.; Gao, Y.; Zhou, Y.; Xu, F.; Deng, X. Coordinative Optimization for Wind Farms Considering Improved Fatigue Load Index. *Int. J. Electr. Power Energy Syst.* **2024**, *162*, 110282. [\[CrossRef\]](#)
25. Huang, Z.; Yang, X.; Hu, S.; Guo, Y.; Wang, Y.; Liu, X.; Wang, Y.; Liang, W.; Yang, J. An Optimal Active Power Allocation Method for Wind Farms Considering Unit Fatigue Load. *Sustainability* **2025**, *17*, 9189. [\[CrossRef\]](#)
26. Zhao, H.; Wu, Q.; Huang, S.; Shahidehpour, M.; Guo, Q.; Sun, H. Fatigue Load Sensitivity-Based Optimal Active Power Dispatch for Wind Farms. *IEEE Trans. Sustain. Energy* **2017**, *8*, 1247–1259. [\[CrossRef\]](#)
27. Guo, J.; Liu, M.; Fang, Z.; Chen, W.; Pan, X.; Xiao, L. Tower Loads Characteristics of a Semi-Submersible Floating Wind Turbine: An Experimental Study. *Ocean Eng.* **2024**, *311*, 118967. [\[CrossRef\]](#)
28. Ge, J.; Sun, Y.; Zhou, S. Fatigue Life Estimation under Multiaxial Random Loading by Means of the Equivalent Lemaitre Stress and Multiaxial S–N Curve Methods. *Int. J. Fatigue* **2015**, *79*, 65–74. [\[CrossRef\]](#)
29. Tian, H.; Soltani, M.N. Vibration Suppression in Wind Turbines via a New Damping System: Characteristics and Performance Evaluation. *IEEE Trans. Ind. Appl.* **2026**, *62*, 1296–1310. [\[CrossRef\]](#)
30. Lara, M.; Mulders, S.P.; Van Wingerden, J.-W.; Vázquez, F.; Garrido, J. Analysis of Adaptive Individual Pitch Control Schemes for Blade Fatigue Load Reduction on a 15 MW Wind Turbine. *Appl. Sci.* **2023**, *14*, 183. [\[CrossRef\]](#)
31. Jonkman, B.; Mudafort, R.M.; Platt, A.; Branlard, E.; Sprague, M.; Ross, H.; Jonkman, HaymanConsulting; Hall, M.; Slaughter, D.; et al. *OpenFAST/Openfast: V3.5.0*; Zenodo: Geneva, Switzerland, 2023. [\[CrossRef\]](#)
32. Abbas, N.J.; Zalkind, D.S.; Pao, L.; Wright, A. A Reference Open-Source Controller for Fixed and Floating Offshore Wind Turbines. *Wind Energy Sci.* **2022**, *7*, 53–73. [\[CrossRef\]](#)
33. Jonkman, J.; Butterfield, S.; Musial, W.; Scott, G. *Definition of a 5-MW Reference Wind Turbine for Offshore System Development*; Technical Report; National Renewable Energy Lab. (NREL): Golden, CO, USA, 2009.
34. Jin, Y.; Wu, D.; Ju, P.; Rehtanz, C.; Wu, F.; Pan, X. Modeling of Wind Speeds Inside a Wind Farm with Application to Wind Farm Aggregate Modeling Considering LVRT Characteristic. *IEEE Trans. Energy Convers.* **2020**, *35*, 508–519. [\[CrossRef\]](#)
35. Gasparis, G.; Lio, W.H.; Meng, F. Surrogate Models for Wind Turbine Electrical Power and Fatigue Loads in Wind Farm. *Energies* **2020**, *13*, 6360. [\[CrossRef\]](#)

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.